Today's Agenda

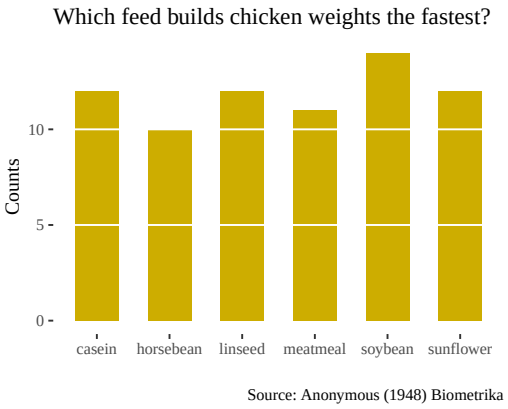
Data Project 1: Due Sunday (Sep 19)

Get ready for class:

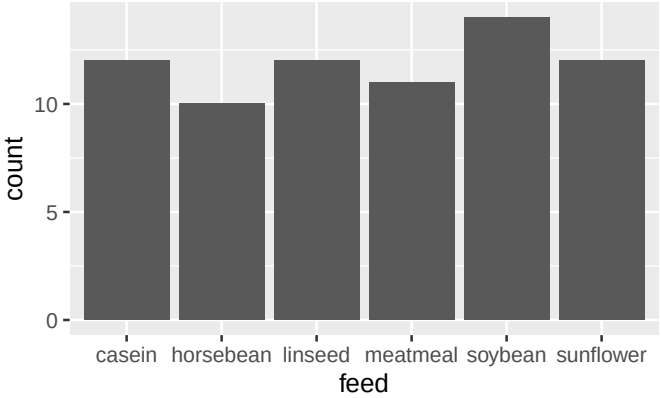
- install.packages("rworldmap")

Polishing Visualizations

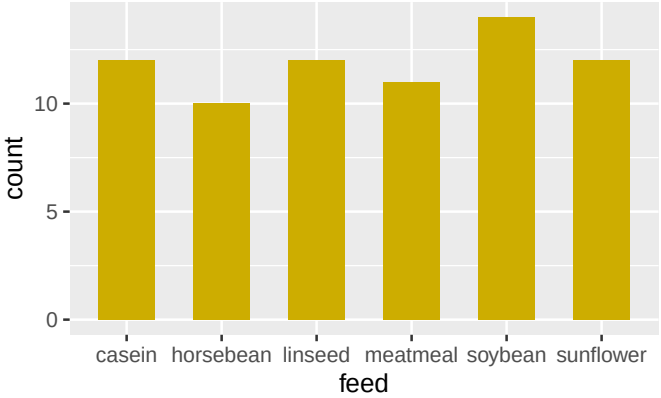
1. Informative titles
2. Figure labels
3. Clean axis labels
4. Source info
5. No chart junk



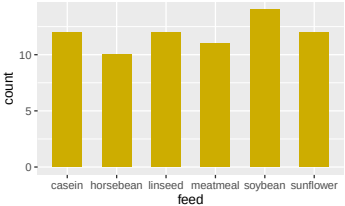
```r
# Map the tidy data
p <- ggplot(data = chickwts, aes(x = feed))

# Visualize the default plot
p +
    geom_bar()
```



```r
# Color in the bars and narrow the width
p +
    geom_bar(fill = "gold3", width = .6)
```

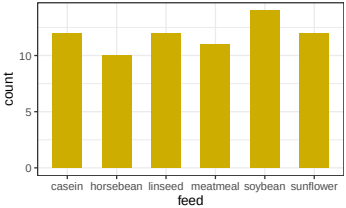


```r
p +
    geom_bar(fill = "gold3", width = .6) +
    theme_gray()
```

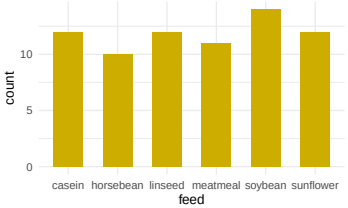



```r
p +
    geom_bar(fill = "gold3", width = .6) +
    theme_classic()
```

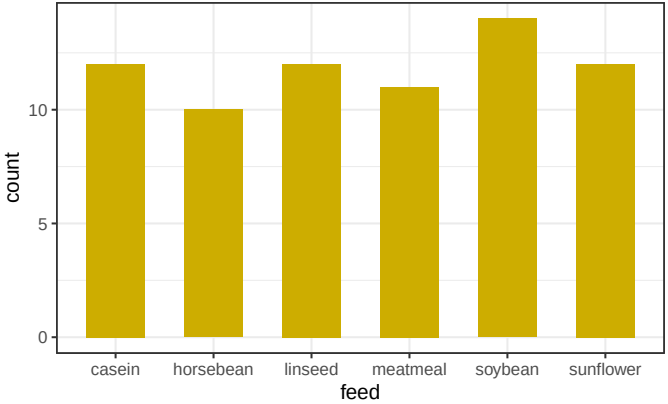
```r
p +
    geom_bar(fill = "gold3", width = .6) +
    theme_bw()
```



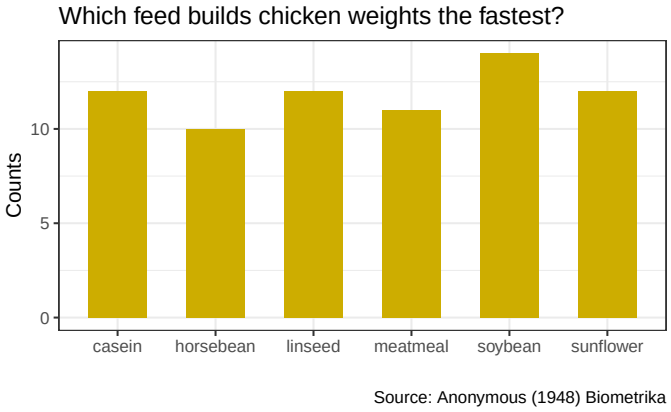

```r
p +
    geom_bar(fill = "gold3", width = .6) +
    theme_minimal()
```



```r
# Let's try theme bw
p +
    geom_bar(fill = "gold3", width = .6) +
    theme_bw()
```

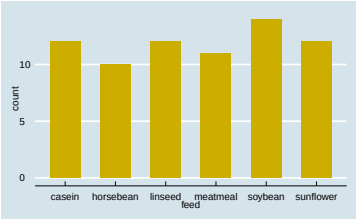


```r
# Modify the axes, captions and titles
p +
    geom_bar(fill = "gold3", width = .6) +
    theme_bw() +
    labs(x = "", y = "Counts", caption = "Source: Anonymous (1948) Biometrika",
         title = "Which feed builds chicken weights the fastest?")
```

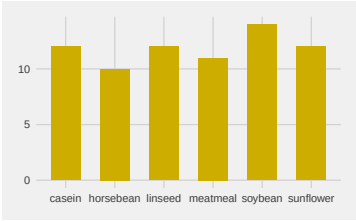


For the brave, we can get crazier...

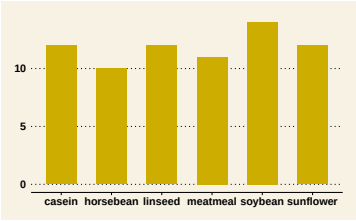
```r
p +
    geom_bar(fill = "gold3", width = .6) +
    ggthemes::theme_economist()
```



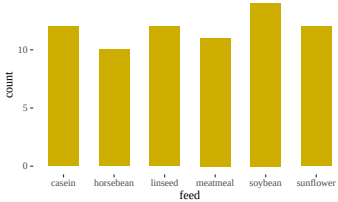

```r
p +
    geom_bar(fill = "gold3", width = .6) +
    ggthemes::theme_fivethirtyeight()
```



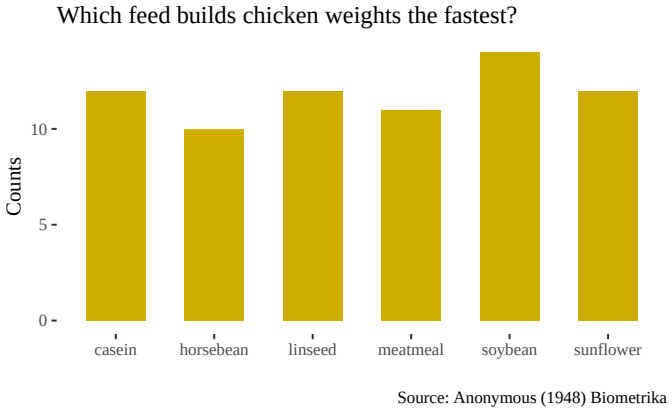

```r
p +
    geom_bar(fill = "gold3", width = .6) +
    ggthemes::theme_wsj()
```



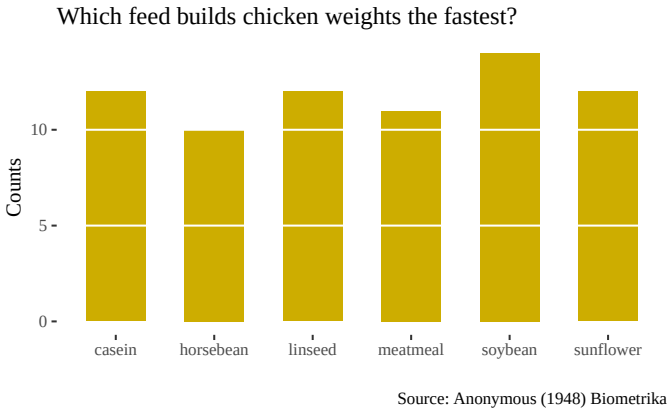

```r
p +
    geom_bar(fill = "gold3", width = .6) +
    ggthemes::theme_tufte()
```



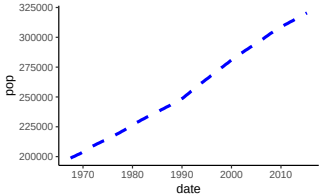
```r
## Fancier themes
p +
    geom_bar(fill = "gold3", width = .6) +
    ggthemes::theme_tufte() +
    labs(x = "", y = "Counts", caption = "Source: Anonymous (1948) Biometrika",
         title = "Which feed builds chicken weights the fastest?")
```

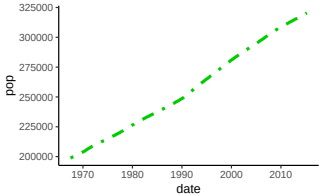


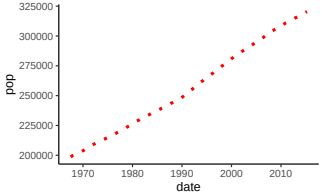
```r
p +
    geom_bar(fill = "gold3", width = .6) +
    ggthemes::theme_tufte() +
    labs(x = "", y = "Counts", caption = "Source: Anonymous (1948) Biometrika",
         title = "Which feed builds chicken weights the fastest?") +
    geom_hline(yintercept = c(5, 10), color = "white")
```

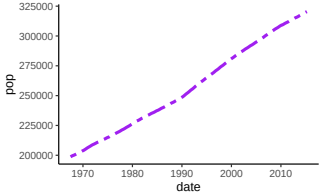


```r
## Customizing a line
## Line types: 'solid', 'dashed', 'dotted', 'dotdash', 'longdash', 'tw
ggplot(data = economics, aes(x = date, y = pop)) +
    geom_line(size = 1.3, color = "blue", linetype = "dashed")
```

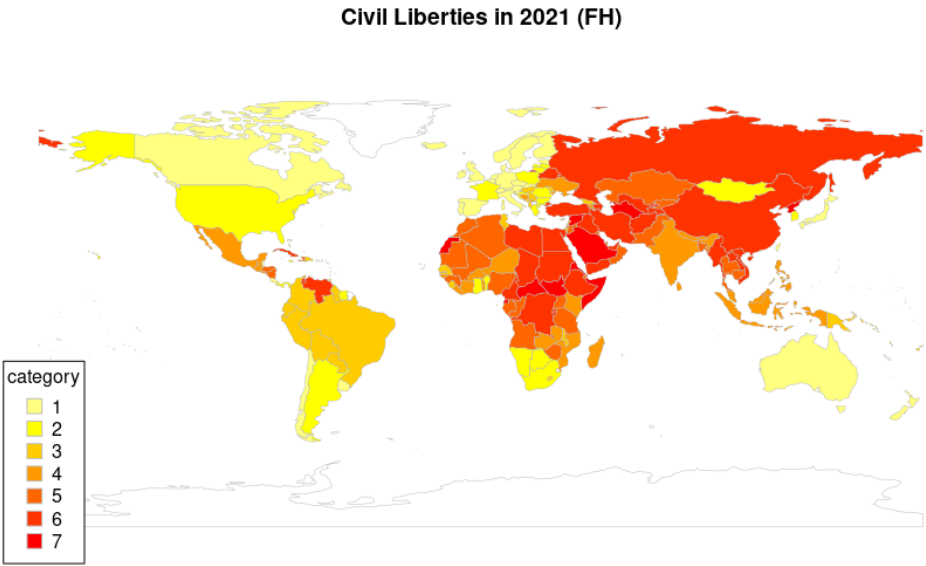








End of Class: Let's make a map!



Tuesday: Working on Paper 1

Prepare a report **introducing and analyzing** our first data project. What are the **strengths and weaknesses** of this project for measuring regime type? Support all claims with evidence.

- Section 1: Introducing the Freedom House Data Project
- Section 2: Analyzing the Methodology
- Section 3: Analyzing the World in 2021

Brainstorming Section 4: Ideas to Test?

Use the data to present and analyze the trends in the world across time

Categorical

- PR rating
- CL rating
- Status

Numerical

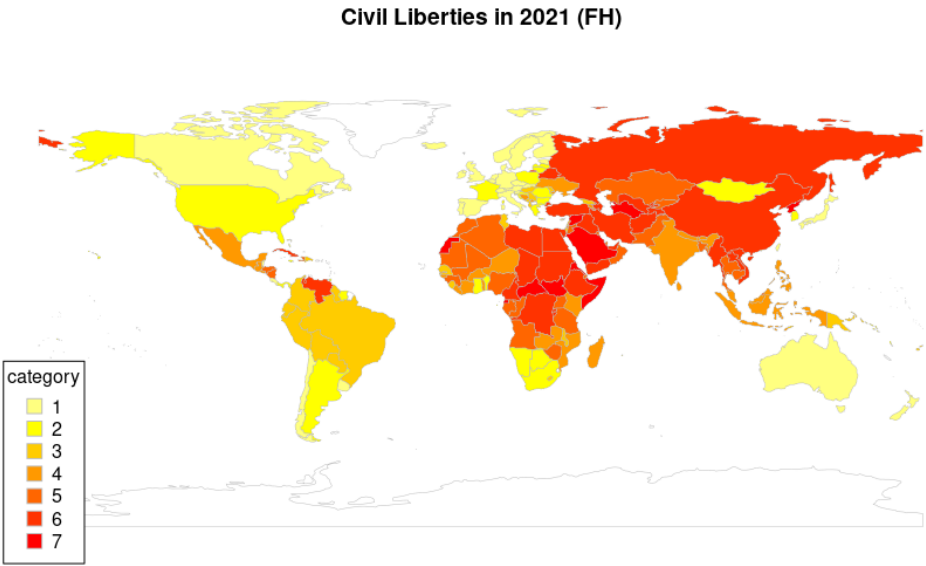
- Total
- PR
- CL

5) Conclusions

Key takeaways

Based on your analyses how can we best use this data project to understand the world?

Let's make a map!



Let's make a map!

1. Load the `rworldmap` package
2. Pick your per country data
3. Link your per country data to the country names in the map
4. Output the map

Let's make a map!

```r
## Install the package (one time)
install.packages("rworldmap") # One time
```

```r
## 1. Load the package
library(rworldmap)
```

Let's make a map!

```r
## 2. Pick your per country data (FH in 2021)
d_2021 <- d |>
            filter(Edition == 2021) |>
            select("Country/Territory", "CL rating")
```

Country/Territory	CL rating
Abkhazia	5
Afghanistan	6
Albania	3
Algeria	5
Andorra	1
Angola	5
Antigua and Barbuda	2
Argentina	2

```r
## 1. Load the package
library(rworldmap)

## 2. Pick your per country data (FH in 2021)
d_2021 <- d |>
            filter(Edition == 2021) |>
            select("Country/Territory", "CL rating")

## 3. Link your data to the map
map1 <- joinCountryData2Map(dF = d_2021,
                            joinCode = "NAME",
                            nameJoinColumn = "Country/Territory")

## 4. Ouput your map
mapCountryData(map1, nameColumnToPlot = "CL rating")
```

```r
## Output a categorical variable and customize colors
mapCountryData(map1,
               nameColumnToPlot = "CL rating",
               mapTitle="Civil Liberties in 2021 (FH)",
               catMethod = "categorical",
               colourPalette = "heat")
```

Color Options

1. a vector of valid colours, e.g. =c('red','white','blue')
2. one of "heat", "diverging", "white2Black", "black2White", "topo", "rainbow", "terrain", "negpos8", "negpos9